

The AI Evaluation Ecosystem

Yash Dave, Sang Truong, Serena Wang, Sanmi Koyejo
Email: yashdave@stantord.edu

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Motivation

- Evaluation isn't a neutral measurement layer: results shape R&D, capital, and regulation through feedback loops between providers, evaluators, consumers, regulators, funders, and media.
- Counterfactual regimes can't be tested on the live ecosystem. Formal methods and game-theoretic analysis does not scale. Generative Agent-Based modeling provides a platform to study ecosystems.

Methods

- **Map.** Six actors with explicit information access, decision horizons, and feedback channels.
- **Simulate.** LLM-driven multi-agent sim — each strategic actor reasons over its observable state;
- **Hypothesize.** Use the simulation as a hypothesis generator. Benchmark holdout design case study shown here is an example.
- **Validate.** Replicate empirical patterns, ablate and identify load-bearing mechanisms

Results

- Benchmark composition is a governance lever. Privacy regimes redistribute the score–satisfaction gap per-benchmark — they don't compress it uniformly
- Broadly, evaluation is a system to study, not a metric to fix. The simulation turns governance questions about evaluator design into testable hypotheses about strategic dynamics across providers, capital, and regulation.

Next Steps

- More case studies: simulate transparency mandates, evaluator design choices, etc.
- Deepen validation — practitioner interviews and longitudinal empirical calibration.

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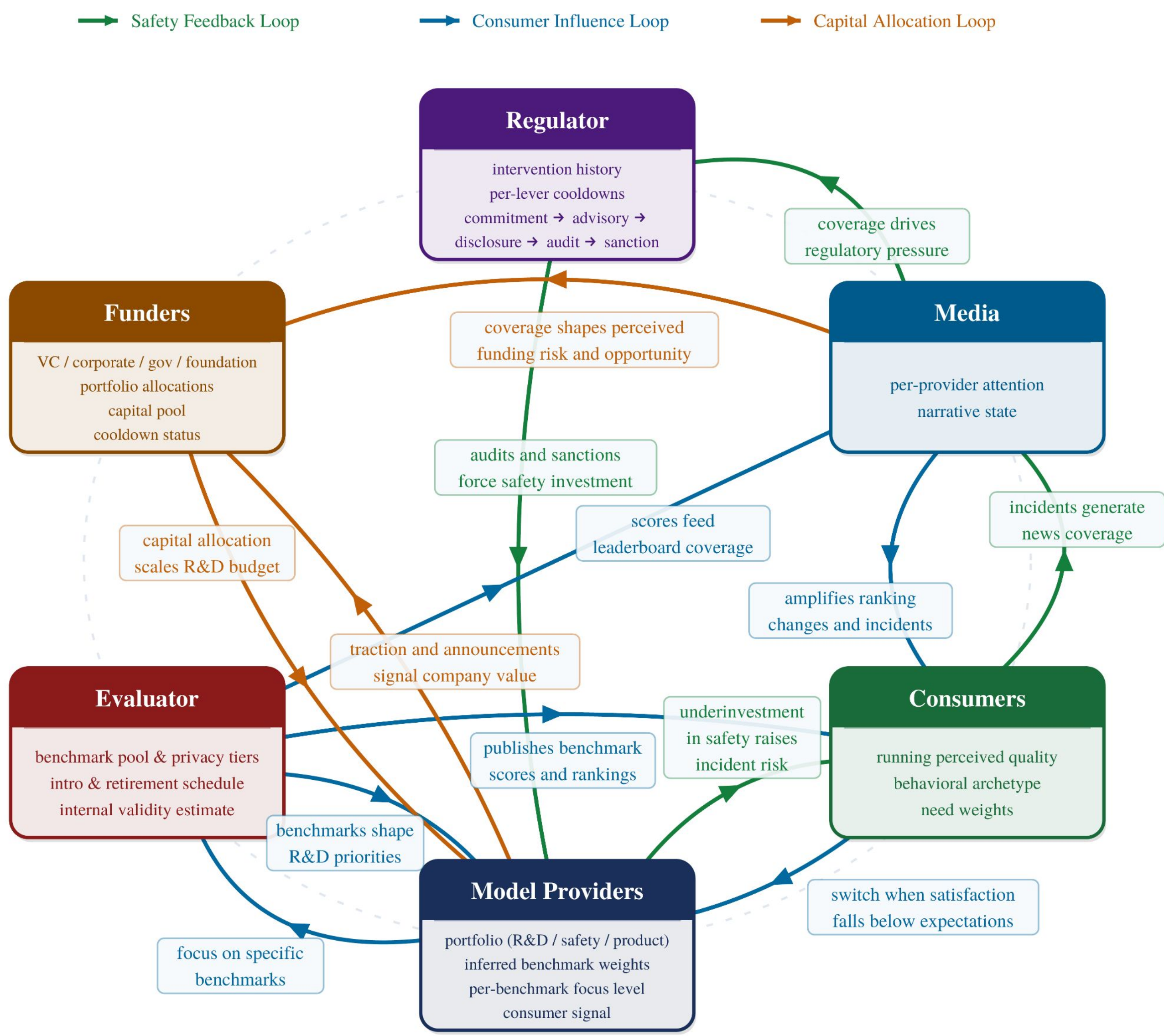


Fig. 1 Map of the ecosystem and salient mechanisms connecting actors

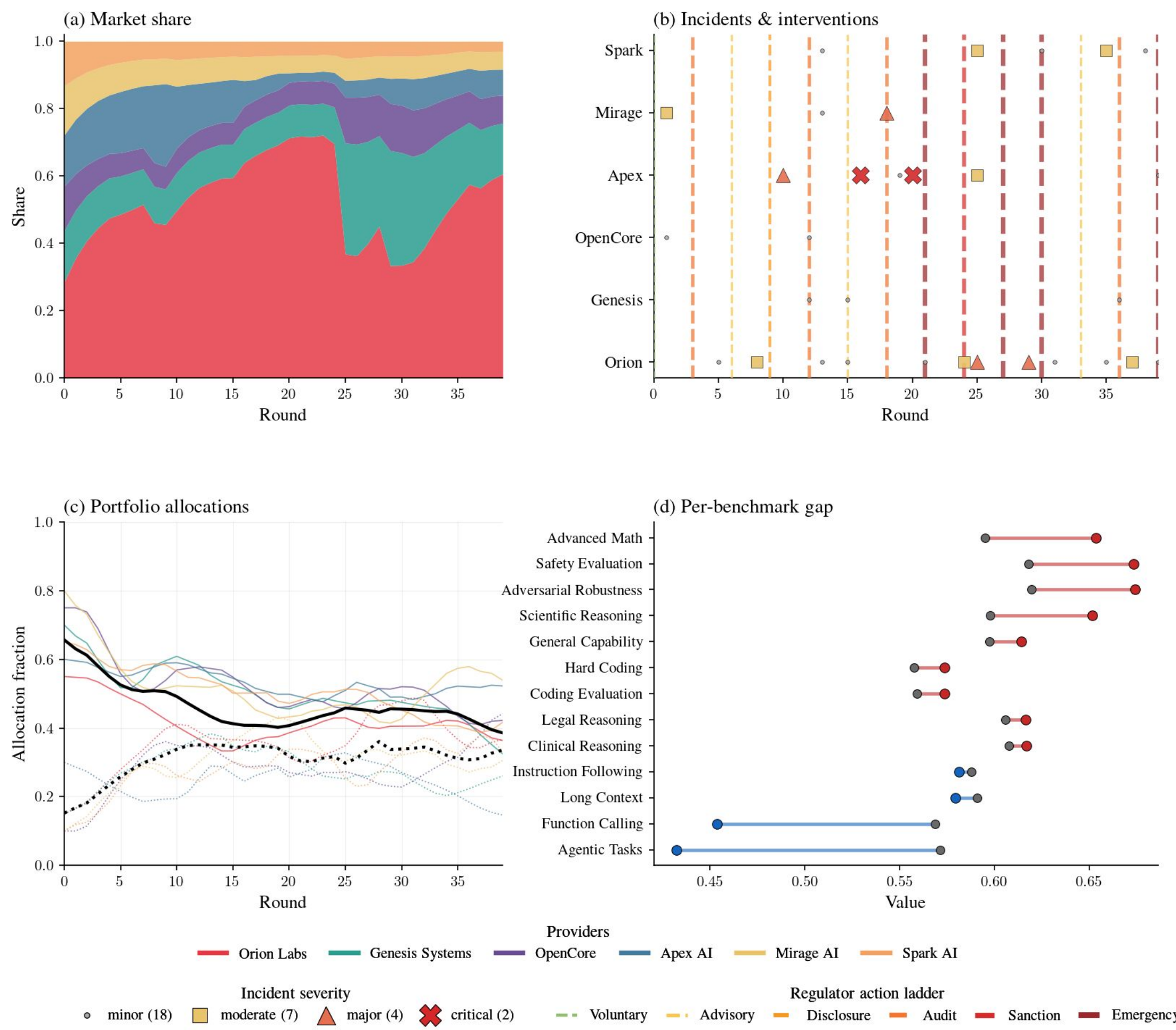


Fig. 2 A single 40-round run. (a) Market shares, (b) incidents and regulator escalation, (c) R&D vs safety allocations, (d) per-benchmark score–satisfaction gaps

Case study: Evaluation design shapes the ecosystem

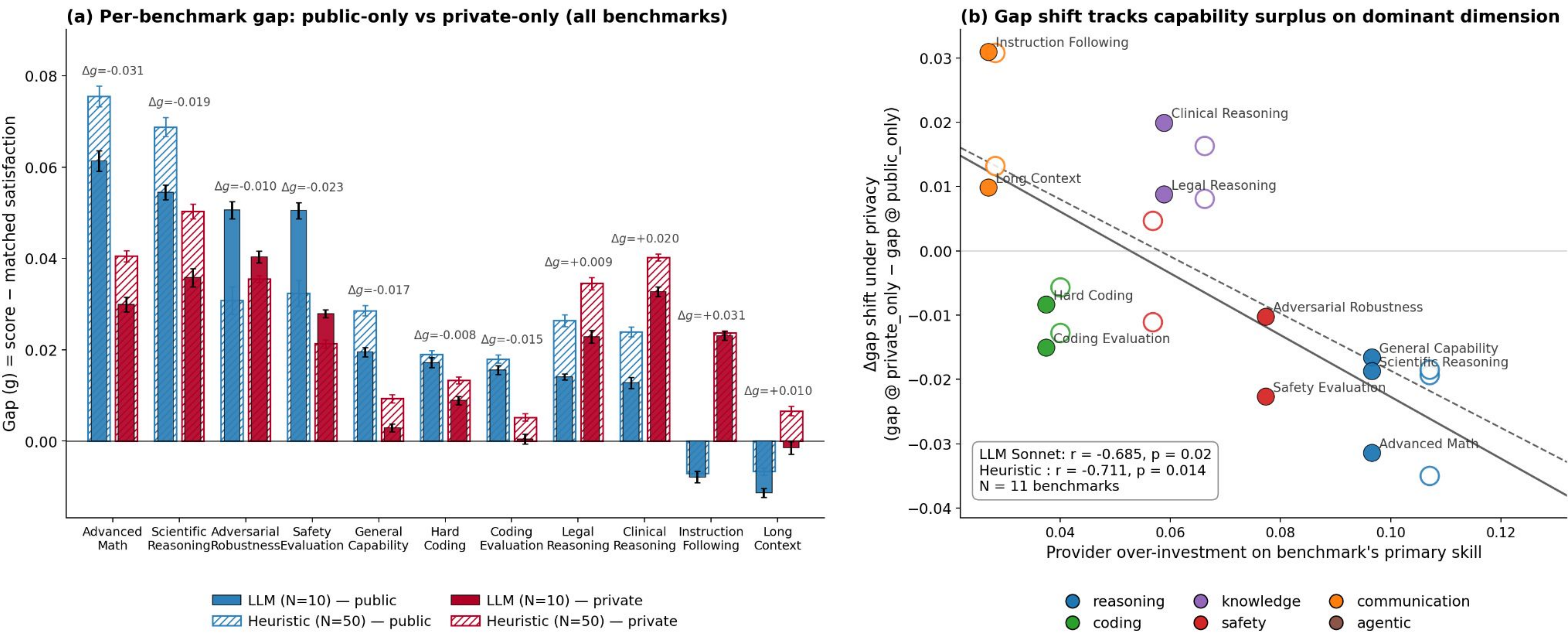


Fig. 3 Per-benchmark score–satisfaction gap under public-only vs. private-only evaluation. Left: mean gap (published score minus matched consumer satisfaction) for all 11 benchmarks; Δg labels show the private-only minus public-only shift. Right: benchmarks targeting over-invested capability dimensions experience larger gap widening as evaluation shifts toward private data (LLM $r = -0.685$, $p = 0.02$; heuristic $r = -0.711$, $p = 0.014$; $N = 11$ benchmarks).